

Designing Constitutional Review Institutions by Simulation

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Abstract

Constitutional review institutions face a persistent design problem: courts need enough independence to protect rights and legal continuity, but enough accountability to avoid becoming an unreviewable partisan veto point. This paper introduces a separate computational simulator for comparing supreme-court and constitutional-review designs. The model varies appointment methods, court size, term limits, removal standards, recusal rules, emergency docket procedures, voting thresholds, opinion-coalition behavior, panel versus en banc review, dual or cross-checking courts, constitutional councils, legislative override rules, and independence/accountability trade-offs. The current version adds docket subtypes, vacancy and replacement dynamics, emergency procedure stages, and override outcomes. A data-import boundary allows legislative simulator outputs to become constitutional dockets without making this project depend on the legislative simulator source tree.

1 Motivation

The motivating question is not whether one court design is ideal in isolation. It is whether institutional designs can be compared under shared docket assumptions and explicit value tradeoffs. A court can be independent while losing democratic legitimacy, responsive while failing rights protection, stable while entrenching bad doctrine, or active while creating constitutional conflict. The simulator is built to expose those tensions rather than collapse them into a single institutional label.

The current model should be read as a design-search scaffold. It does not claim to forecast any real supreme court. It creates synthetic constitutional cases, routes them through institutional designs, and reports comparative indicators. The v1 campaign is therefore useful for identifying which mechanics deserve deeper calibration: appointment selection, emergency-docket restraints, cross-court review, council warnings, override channels, recusal discipline, and replacement timing.

2 Model

Each generated world contains a pool of potential justices, a constitutional docket, and a legislative-output profile. Cases vary by legal ambiguity, rights burden, democratic mandate, partisan salience, emergency pressure, requested emergency relief, legislative quality, constitutional-conflict potential, public attention, and override pressure. The docket separates facial challenges, as-applied challenges, election disputes, emergency stay applications, executive-power disputes, administrative-law challenges, and rights claims. Justices vary by ideology, independence, rights sensitivity, accountability pressure, institutionalism, partisan loyalty, and emergency deference.

Each scenario supplies a court design. The court design selects reviewers, applies recusal rules, routes cases to full-court or panel review, determines whether emergency relief occurs through

an application-only path, response window, reasoned temporary stay, shadow stay, merits acceleration, or expiration without relief, applies a voting threshold, optionally invokes cross-checking or council review, and optionally allows legislative or popular override. Override attempts can fail, succeed through ordinary supermajority action, proceed through delayed reenactment or referendum, be blocked by a rights carveout, or recur as a repeated override. Decisions update a simple court state that tracks precedent stability and conflict load over time. Court rosters also change over simulated time through term-limit schedules, retention pressure, life-tenure attrition, and appointment-method-specific replacement selection.

3 Legislative Input Boundary

This project remains separate from the legislative simulator. Legislative outputs are imported as CSV data and summarized into a neutral profile with enacted volume, legal quality, weak-mandate rate, rights risk, partisan skew, volatility, public legitimacy, and override pressure. This allows laws generated under legislative simulations to become inputs for constitutional review without coupling Java packages or build systems.

The v1 campaign was run with the adjacent congressional simulator’s main campaign CSV as an imported input. The importer summarized that file as enacted volume 0.324, legal quality 0.617, weak-mandate rate 0.174, rights risk 0.106, partisan skew 0.233, volatility 0.121, and public legitimacy 0.550.

4 Metrics

The headline metrics are legal stability, rights protection, partisan alignment, shadow-docket abuse, legitimacy, reversal rate, constitutional conflict, and democratic responsiveness. Higher legal stability, rights protection, legitimacy, and democratic responsiveness are generally better. Lower partisan alignment, shadow-docket abuse, reversal rate, constitutional conflict, and administrative cost are generally better. Invalidation, emergency order, merits acceleration, replacement, recusal, concurrence, dissent, en banc, council warning, cross-check disagreement, override-attempt, and override-success rates are diagnostic rather than automatically good or bad.

The reported directional score is a reading aid. It averages a stability/rights score, a legitimacy/control score, and administrative feasibility. It should not be treated as a final constitutional ranking.

5 V1 Campaign

The current campaign compares thirteen institutional scenarios across fifteen assumption cases: the original broad cases plus sensitivity sweeps for low and extreme appointment capture, low and extreme emergency pressure, low and extreme rights risk, weak-mandate legislation, and strong-mandate legislation. The reported run used 80 runs per case, 64 cases per run, seed 20260501, and the imported congressional simulator campaign CSV.

6 Early Interpretation

The v1 results preserve the main v0 pattern while making the mechanics easier to inspect. Supermajority invalidation remains the top directional-score scenario under the current score, mostly because

Scenario	Score	Stability	Rights	Partisan	Shadow	Conflict
60 percent invalidation threshold	0.809	0.950	0.703	0.062	0.043	0.279
18-year staggered terms	0.805	0.934	0.708	0.060	0.060	0.280
Stylized current U.S.-like court	0.799	0.926	0.709	0.061	0.219	0.281
No emergency relief without merits review	0.797	0.953	0.702	0.040	0.001	0.279
Panel with en banc correction	0.797	0.937	0.707	0.071	0.059	0.280
Cross-checking constitutional court	0.759	0.947	0.700	0.025	0.043	0.316
Dual supreme courts	0.741	0.916	0.710	0.024	0.043	0.318

Table 1: Selected v1 campaign averages. Lower values are preferred for partisan alignment, shadow-docket abuse, and conflict. The full CSV and Markdown report are generated under `reports/`.

it combines stability with low administrative cost. The no-emergency-relief-without-merits design nearly eliminates shadow-docket abuse but pays administrative cost through merits acceleration. Cross-checking and dual-court designs reduce partisan alignment, but the current disagreement-resolution model increases constitutional conflict and administrative cost. Replacement dynamics make term-limited and panel systems more active procedurally without automatically dominating life tenure.

7 Remaining Work

The next increments should be calibration and diagnostics rather than more named court designs. The highest-value additions are:

1. calibrate docket mixes against historical court docket categories rather than hand-set synthetic shares;
2. add empirical comparison targets for emergency orders, merits timing, invalidation frequency, and recusal frequency;
3. add mechanism ablations that isolate appointment method, term limit, emergency rule, threshold, override, and cross-checking effects;
4. add seed robustness and uncertainty bands to the v1 campaign tables;
5. separate legal stability into precedent stability, statutory stability, and interbranch compliance;
6. test manipulation cases where legislatures time appointments, flood emergency applications, or tailor overrides to evade rights carveouts.

8 Reproducibility

The simulator is dependency-free Java and is run through `make`. The core commands are `make test`, `make campaign`, and `make paper`. The v1 report artifacts are written under `reports/`. The preserved v0 campaign can still be regenerated with `make campaign-v0`. The paper is built from this file with `make paper`. The software and campaign outputs are treated as the primary artifacts [2]. Imported legislative profiles are optional and come from the separate legislative simulator’s report CSV contract [1].

References

- [1] Congress Institutional Simulator Project. Congress institutional simulator, 2026. Adjacent legislative simulator used as an optional source of imported legislative-output profiles.
- [2] Constitutional Review Simulator Project. Constitutional review simulator, 2026. Local reproducible simulation project and v1 campaign artifacts.